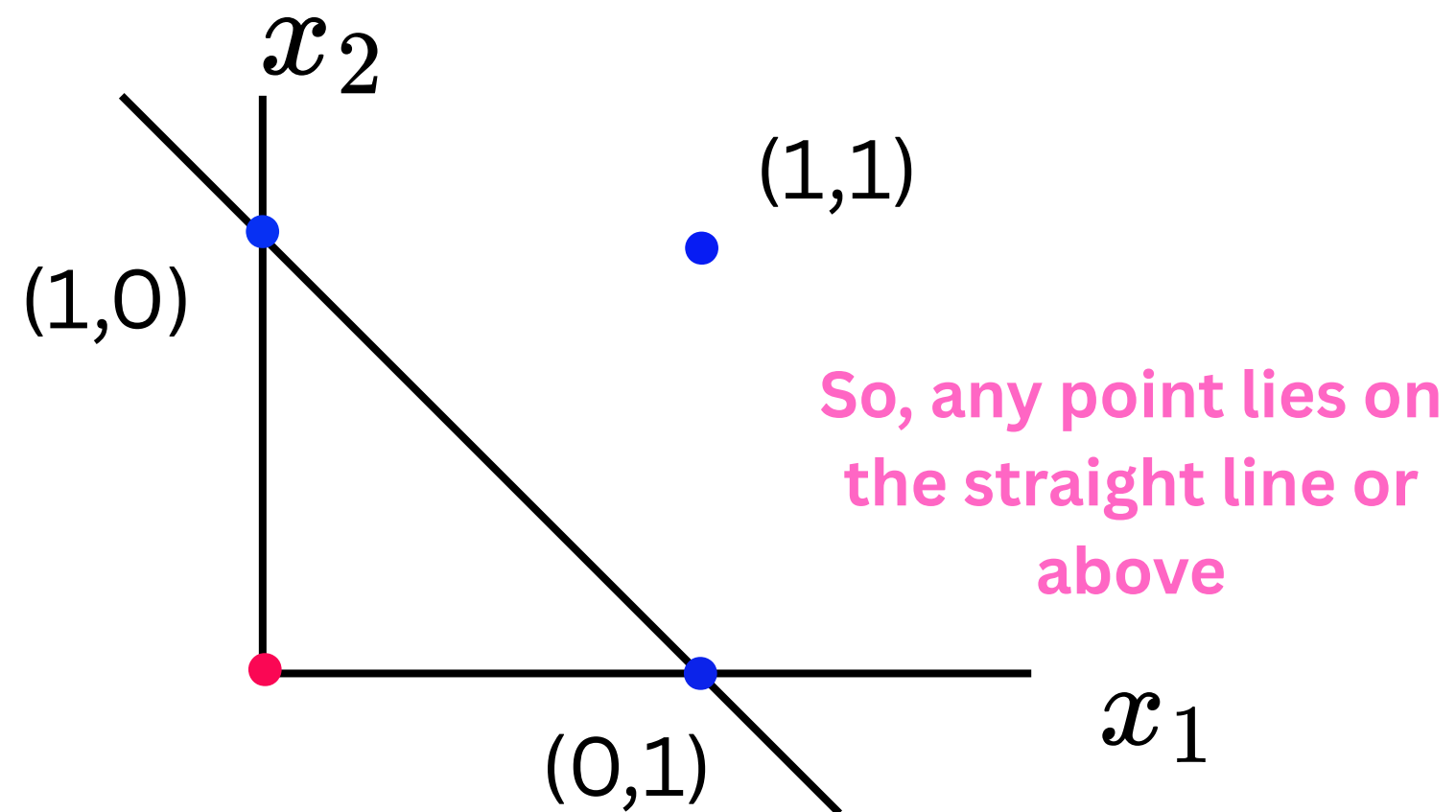




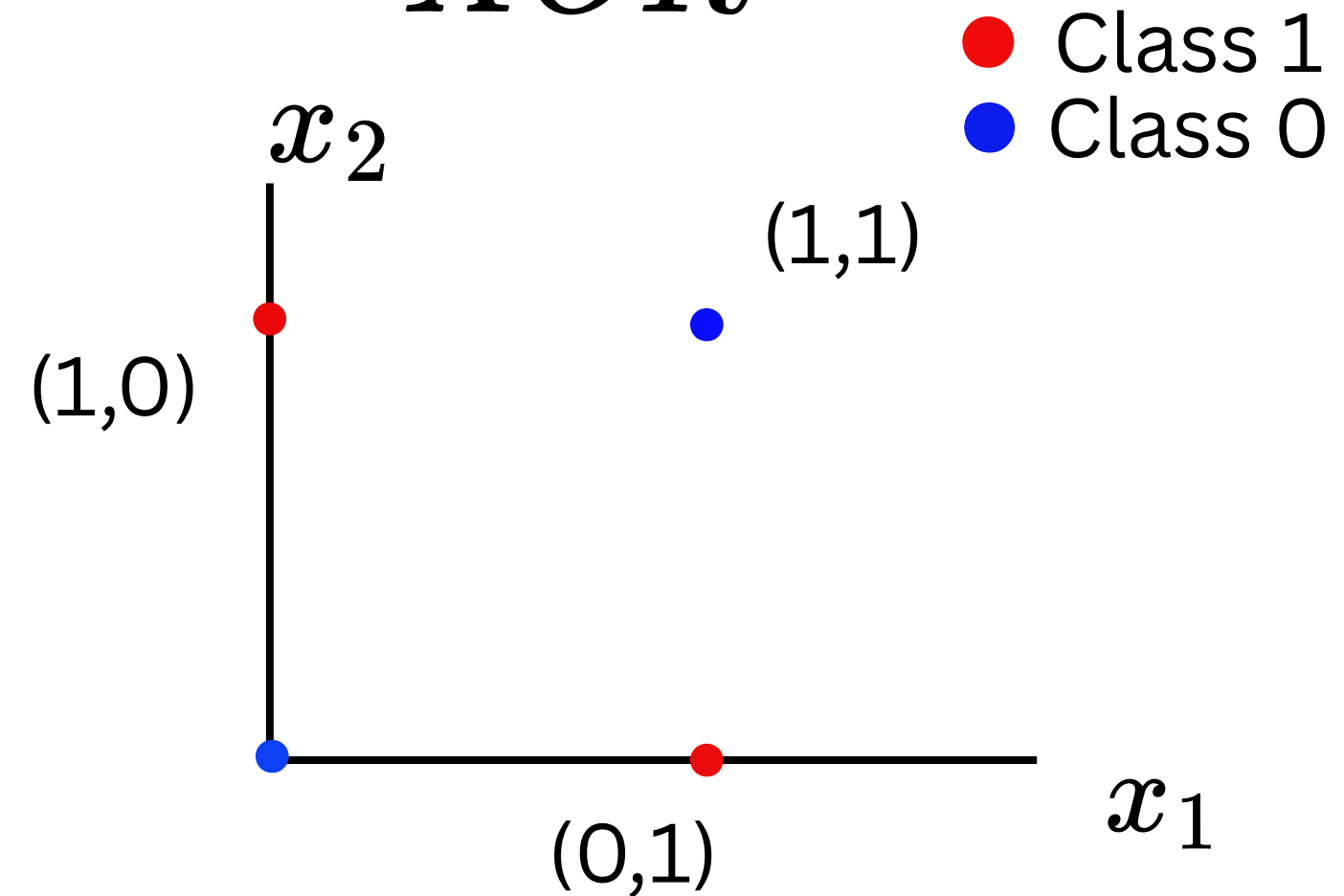
Separability and Non-Separable problem

OR $\theta \geq 1$



- In above scenario we can easily draw a line to separate these two classes

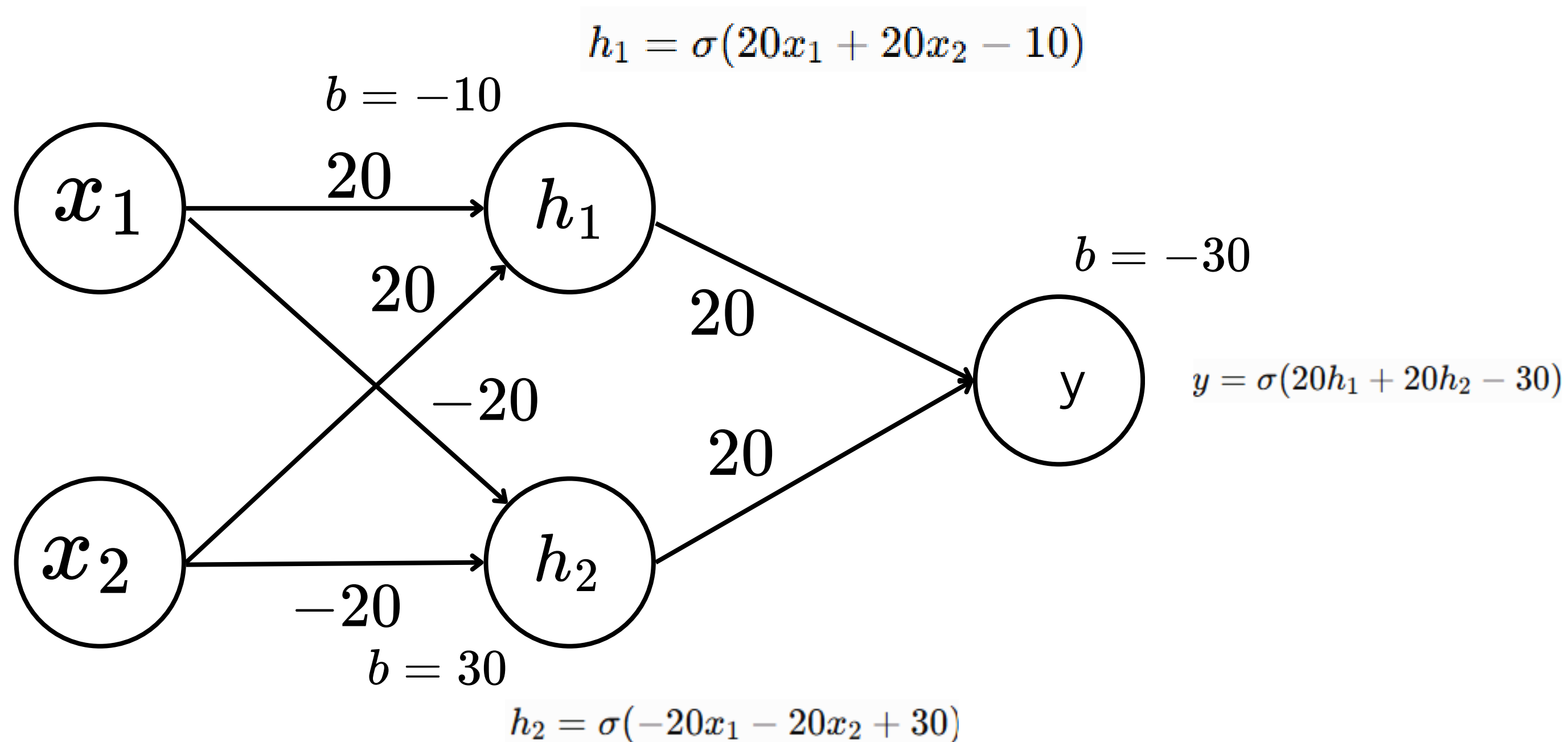
XOR



- In above scenario there is no way that we can draw a line to separate these two classes
- The problem is called XOR problem



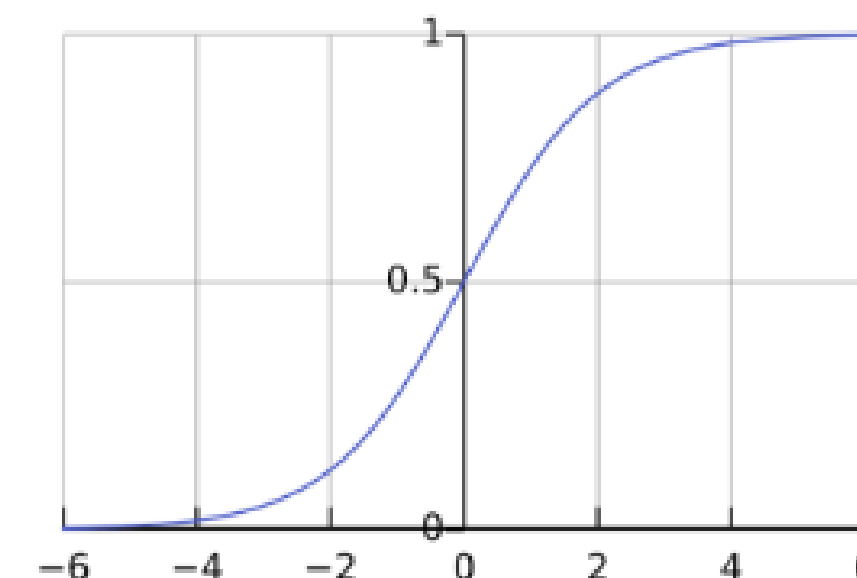
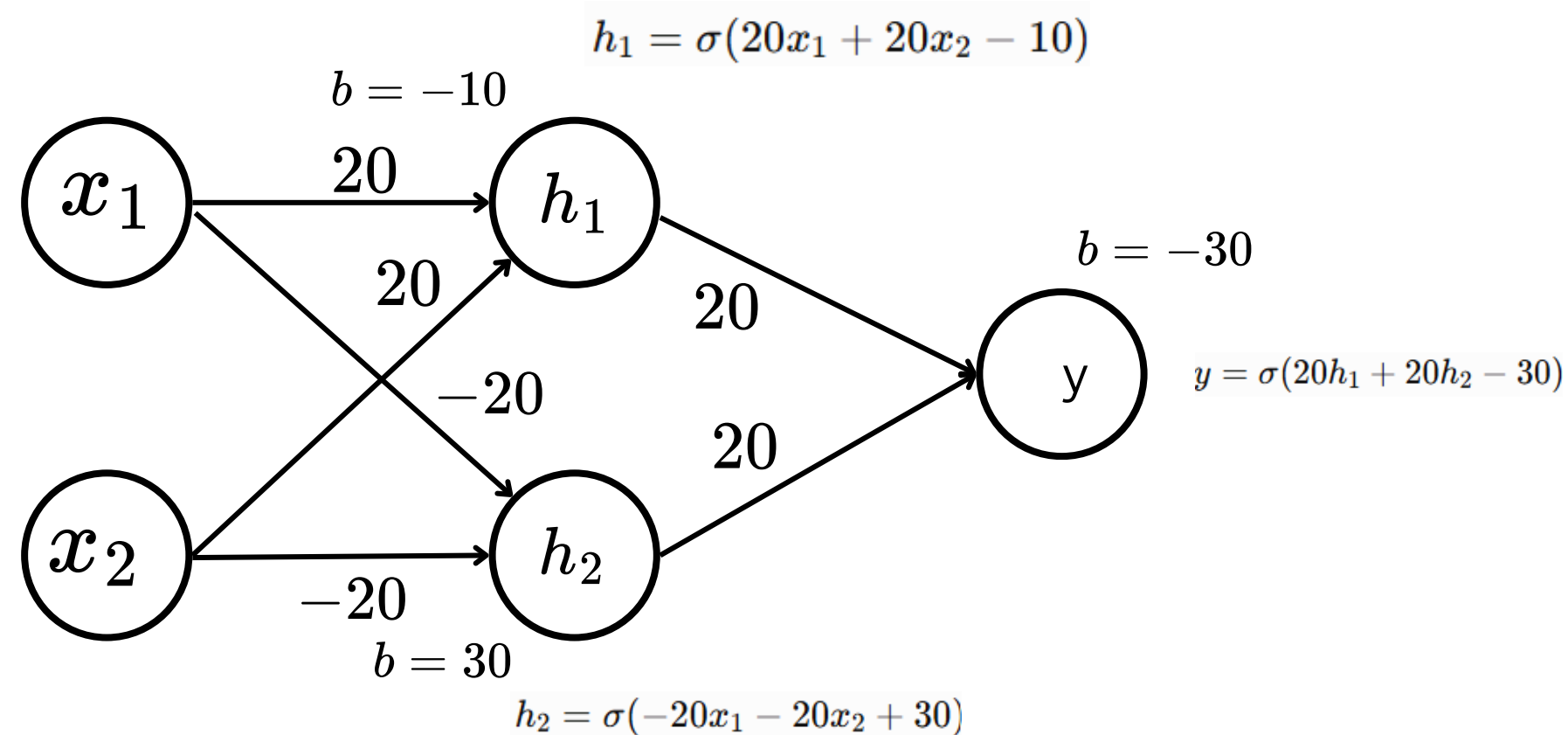
XOR Problem Solution With MLP



An Example Network Set-up to Solve XOR problem



XOR Problem Solution With MLP

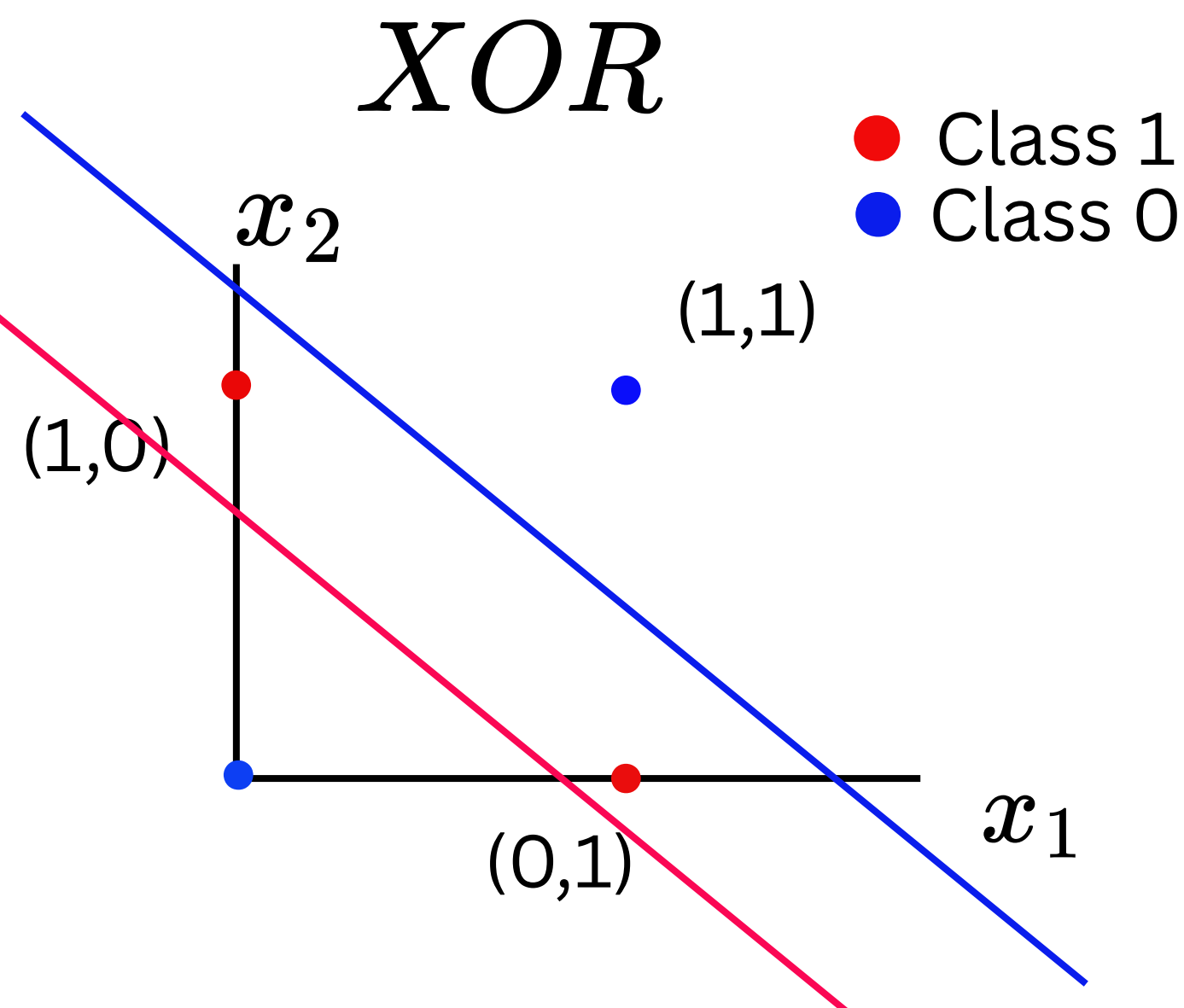
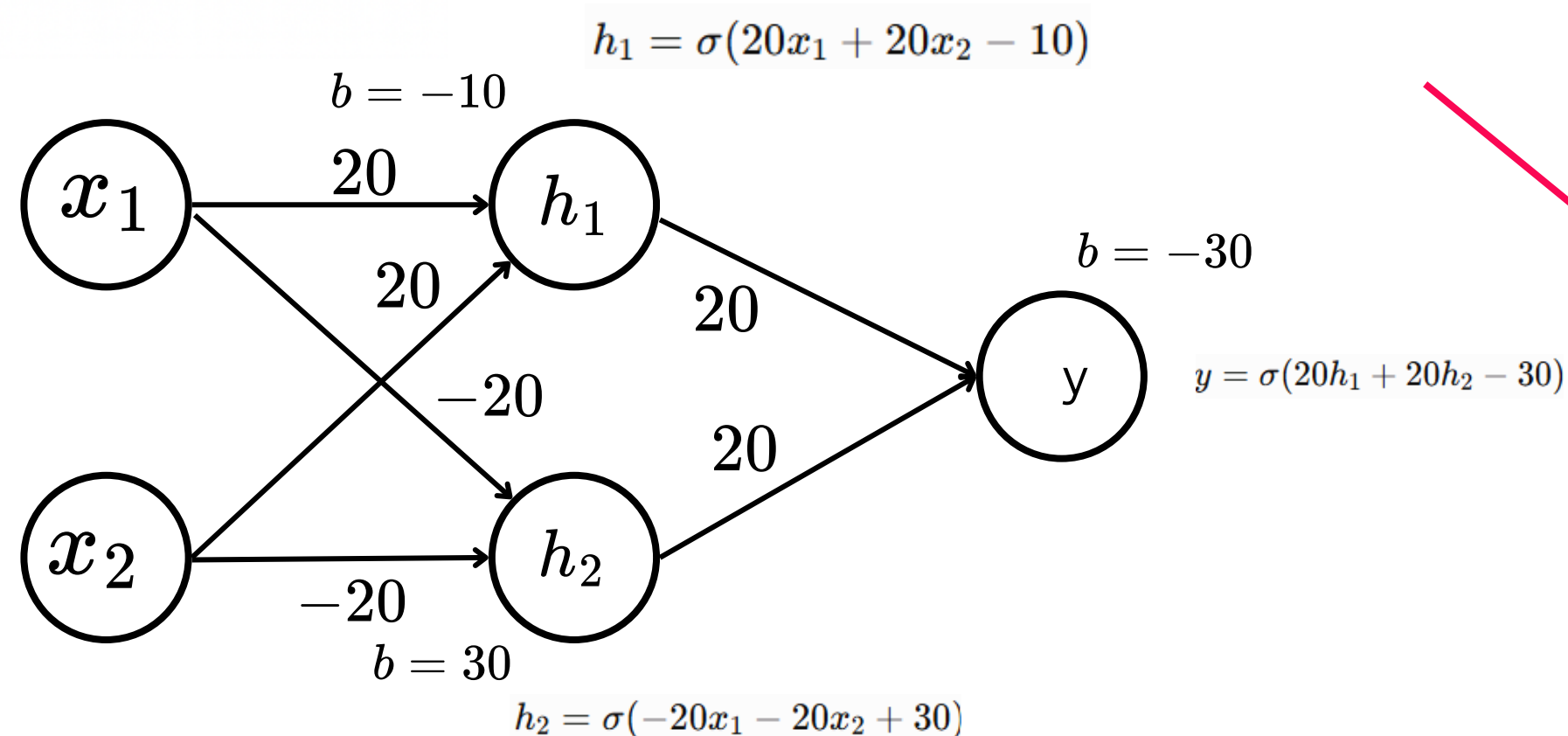


$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

(x_1, x_2)	h_1	h_2	y
(0, 0)	$\sigma(20(0) + 20(0) - 10) \approx 0$	$\sigma(-20(0) - 20(0) + 30) \approx 1$	$\sigma(20(0) + 20(1) - 30) \approx 0$
(0, 1)	$\sigma(20(0) + 20(1) - 10) \approx 1$	$\sigma(-20(0) - 20(1) + 30) \approx 1$	$\sigma(20(1) + 20(1) - 30) \approx 1$
(1, 0)	$\sigma(20(1) + 20(0) - 10) \approx 1$	$\sigma(-20(1) - 20(0) + 30) \approx 1$	$\sigma(20(1) + 20(1) - 30) \approx 1$
(1, 1)	$\sigma(20(1) + 20(1) - 10) \approx 1$	$\sigma(-20(1) - 20(1) + 30) \approx 0$	$\sigma(20(1) + 20(0) - 30) \approx 0$



XOR Problem Solution With MLP



- So, what the MLP is doing, it is creating two hyper planes to segregate the two classes, which was otherwise impossible by single layer perceptron.
- The points between the planes belongs to class 1 and otherwise class 0